Team name	[Your Team Name]
Project title	StudyPulse: Collaborative Lecture-to-Active-Recall Workspace
Selected case	Case 1: Smarter Campus and Learning
Members	[Member A (leader), Member B, Member C, Member D]

Abstract

1. Problem. Undergraduate students taking quantitative and theory-heavy courses struggle to consolidate fragmented learning materials across dense PDF lecture slides, handwritten whiteboard derivations, and fast-paced spoken explanations. When studying for exams individually or in small groups, students spend hours passively re-reading static files rather than engaging in structured active recall, while deaf or hard-of-hearing students face additional barriers following uncaptioned lectures.

2. Intended Users. First- and second-year undergraduate engineering and computer science students studying for midterm and final examinations in study groups of 3 to 5 peers, as well as peer tutors who lead weekly course review sessions.

3. Supporting Observations or Evidence. In a preliminary observation and survey of [N, e.g., 35] undergraduate students at our campus [1], [X, e.g., 28] reported that lecture slides uploaded to the campus LMS lack the detailed explanations spoken in class, and students estimated spending an average of [Y, e.g., 2.5] hours per course manually rewriting notes before exams. Furthermore, we observed that group study notes shared via chat apps quickly become outdated and unindexed.

4. Proposed Solution and Core User Flow. StudyPulse is a collaborative web workspace that transforms raw course materials into synchronized notes, visual concept maps, and spaced-repetition quizzes. The core user flow has four steps: (1) A student signs in and creates a shared course workspace by uploading lecture PDFs, whiteboard photos, or class audio recordings. (2) The system parses documents and aligns audio timestamps to specific slide sections inside a real-time collaborative block editor where group members co-edit notes simultaneously. (3) With one click, students generate an interactive concept mindmap and a deck of review flashcards from their verified notes. (4) Students complete a 10-minute daily active-recall quiz scheduled by a spaced-repetition algorithm, while a group readiness dashboard highlights weak topics for their next study session. The first version focuses on PDF/image slide parsing, collaborative note editing, and flashcard scheduling for two undergraduate courses.

5. How It Differs from Existing Alternatives. General note-taking apps (such as Notion or Google Docs) lack native LaTeX OCR, slide-aligned audio timestamps, and built-in spaced-repetition scheduling, whereas standalone flashcard tools (such as Anki) do not support real-time multiplayer note collaboration. StudyPulse unifies document extraction, live co-editing, and FSRS-based active recall in one browser workflow.

6. Planned Technologies. A responsive Next.js and TypeScript web application backed by Supabase (PostgreSQL, authentication, and real-time sync). Collaborative editing is powered by BlockNote and Yjs CRDTs, visual concept maps by Markmap, and review scheduling by ts-fsrs. A FastAPI Python service uses MarkItDown and LaTeX-OCR for document extraction, WhisperX for timestamped transcription, and the Google Gemini API strictly for grounded question generation from uploaded notes.

7. Expected Benefit and Development Plan. Expected benefit: reduce note-consolidation time by at least 50% and improve post-study quiz scores during a two-week trial with 20 students, measured via

task completion logs and pre/post-quiz scores. Plan: Week 1 document parser and collaborative editor, Week 2 FSRS flashcard engine and mindmap view, Week 3 user testing and refinement. The main risk is OCR inaccuracy on low-resolution whiteboard photos, which we mitigate by allowing inline manual LaTeX editing before quiz generation.

Source Acknowledgments. [1] Preliminary team survey and observation of undergraduate study habits (November 2026).